SUPERSET ID - 6364957 (Sachin Ray)

React Labs Complete Assignment Solutions (Incl. Additional Handsons upto HOL-8)

Lab 1: Setting up React Environment and First Application

Objective

Create a new React Application with the name "myfirstreact" and display "Welcome to the first session of React" as heading.

Prerequisites

- Node.js
- NPM
- Visual Studio Code

Solution:

1. **Installing Create React App globally:**

```
npm install -g create-react-app
```

2. **Creating React Application:**

```
create-react-app myfirstreact
```

3. **Navigating to project folder:**

```
cd myfirstreact
```

4. **Opening in VS Code and modifying App.js:**

```
import React from 'react';
import './App.css';

function App() {
  return (
    <div className="App">
```

```
    <h1>Welcome to the first session of React</h1>
   </div>
  );
}

export default App;
```

5. **Running the application:**

```
npm start
```

Output:

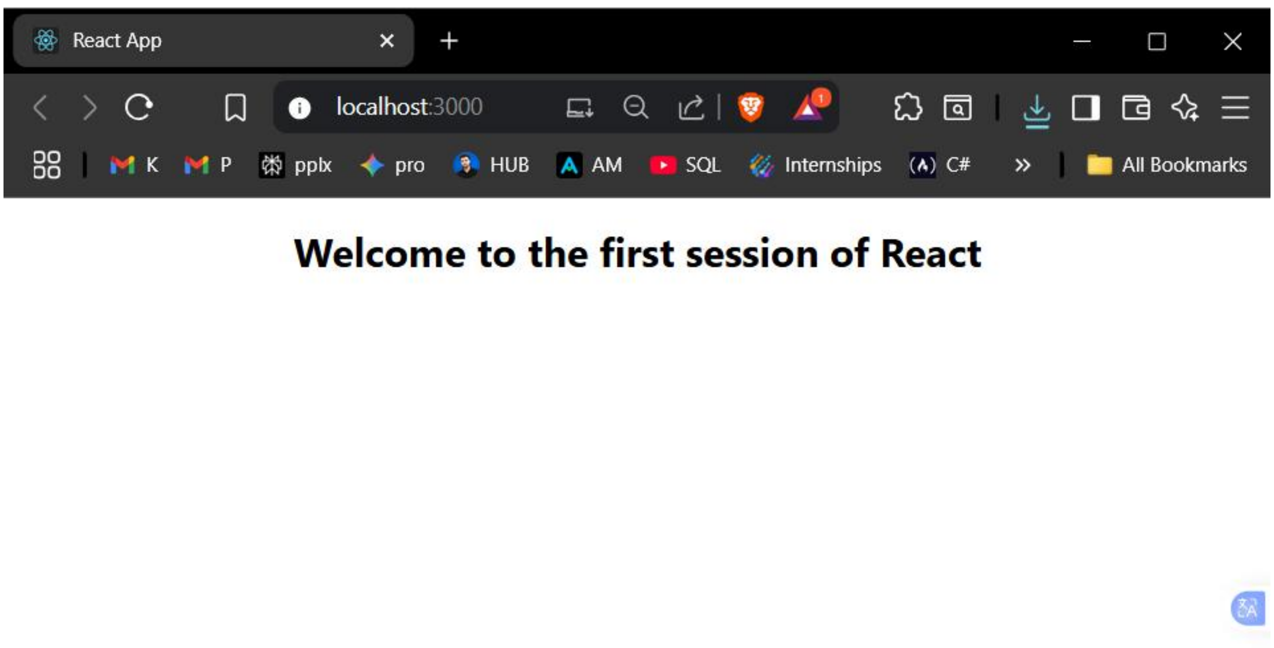

Fig. 1. LAB-1 Output

Lab 2: Creating Multiple Class Components

Objective

Create a Student Management Portal with Home, About, and Contact components.

Solution:

1. **Creatig a React project:**

```
create-react-app student-app
cd StudentApp
```

2. **Creating a Components folder and Home.js:**

```
// src/Components/Home.js
import React, { Component } from 'react';

class Home extends Component {
  render() {
    return (
      <div>
        <h2>Welcome to the Home page of Student Management Portal</h2>
      </div>
    );
  }
}

export default Home;
```

3. **Creating About.js:**

```
// src/Components/About.js
import React, { Component } from 'react';

class About extends Component {
  render() {
    return (
      <div>
        <h2>Welcome to the About page of the Student Management Portal</h2>
      </div>
    );
  }
}

export default About;
```

4. **Creating Contact.js:**

```
// src/Components/Contact.js
import React, { Component } from 'react';
```

```
class Contact extends Component {
  render() {
    return (
      <div>
        <h2>Welcome to the Contact page of the Student Management Portal</h2>
      </div>
    );
  }
}

export default Contact;
```

5. **App.js:**

```
import React from 'react';
import './App.css';
import Home from './Components/Home';
import About from './Components/About';
import Contact from './Components/Contact';

function App() {
  return (
    <div className="App">
      <Home />
      <About />
      <Contact />
    </div>
  );
}

export default App;
```

Output:

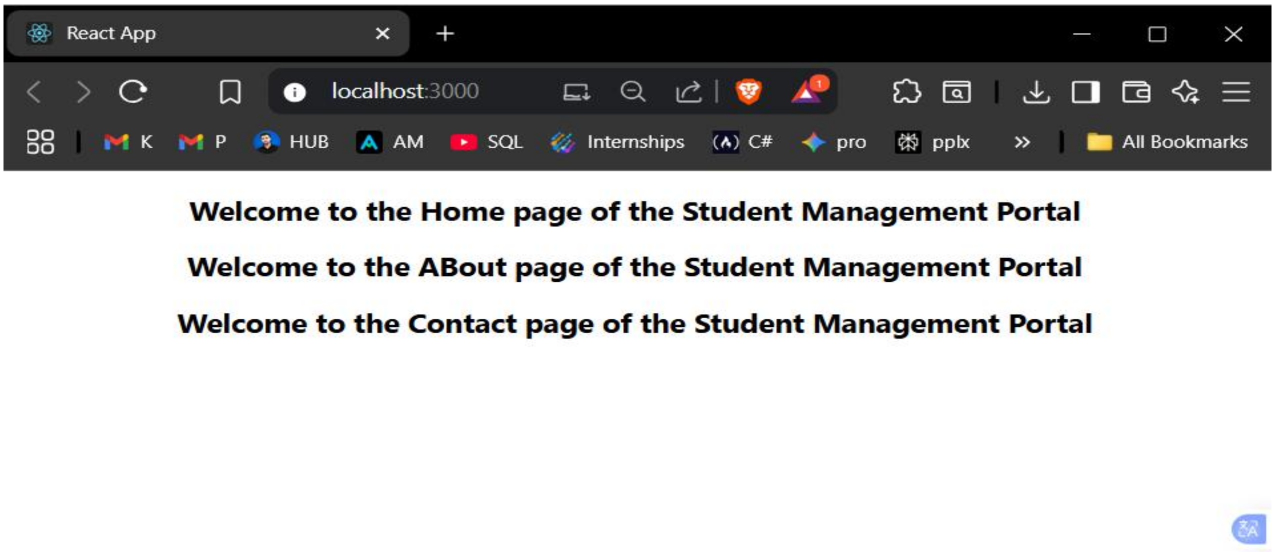

Fig. 2. LAB-2 Output

Lab 3: Function Component with Styling

Objective

Create a score calculator app with functional component and styling.

Solution:

1. **Creating the React project:**

```
create-react-app scorecalculatorapp
cd scorecalculatorapp
```

2. **Creating CalculateScore.js:**

```
// src/Components/CalculateScore.js
import React from 'react';
import '../Stylesheets/mystyle.css';

const CalculateScore = ({ name, school, total, goal }) => {
  const average = (total / goal) * 100;

  return (
    <div className="score-container">
      <h2>Score Calculator</h2>
      <div className="student-info">
        <p><strong>Name:</strong> {name}</p>
        <p><strong>School:</strong> {school}</p>
        <p><strong>Total Score:</strong> {total}</p>
        <p><strong>Goal:</strong> {goal}</p>
        <p><strong>Average Score:</strong> {average.toFixed(2)}%</p>
      </div>
    </div>
  );
};

export default CalculateScore;
```

3. **Creating mystyle.css:**

```
/* src/Stylesheets/mystyle.css */
.score-container {
  max-width: 400px;
  margin: 20px auto;
  padding: 20px;
  border: 2px solid #333;
  border-radius: 10px;
  background-color: #f9f9f9;
  text-align: center;
}

.student-info {
  background-color: white;
  padding: 15px;
  border-radius: 5px;
  margin-top: 10px;
}

.student-info p {
  margin: 10px 0;
```

```
  font-size: 16px;
}

h2 {
  color: #333;
  margin-bottom: 20px;
}
```

4. **App.js:**

```
import React from 'react';

import './App.css';

import CalculateScore from './Components/CalculateScore';


function App() {

  return (

    <div className="App">

      <CalculateScore

        name="Sachin Ray"

        school="Kalinga Institute of Industrial Technology"

        total={93}

        goal={100}

      />

    </div>

  );

}

export default App;
```

Output:

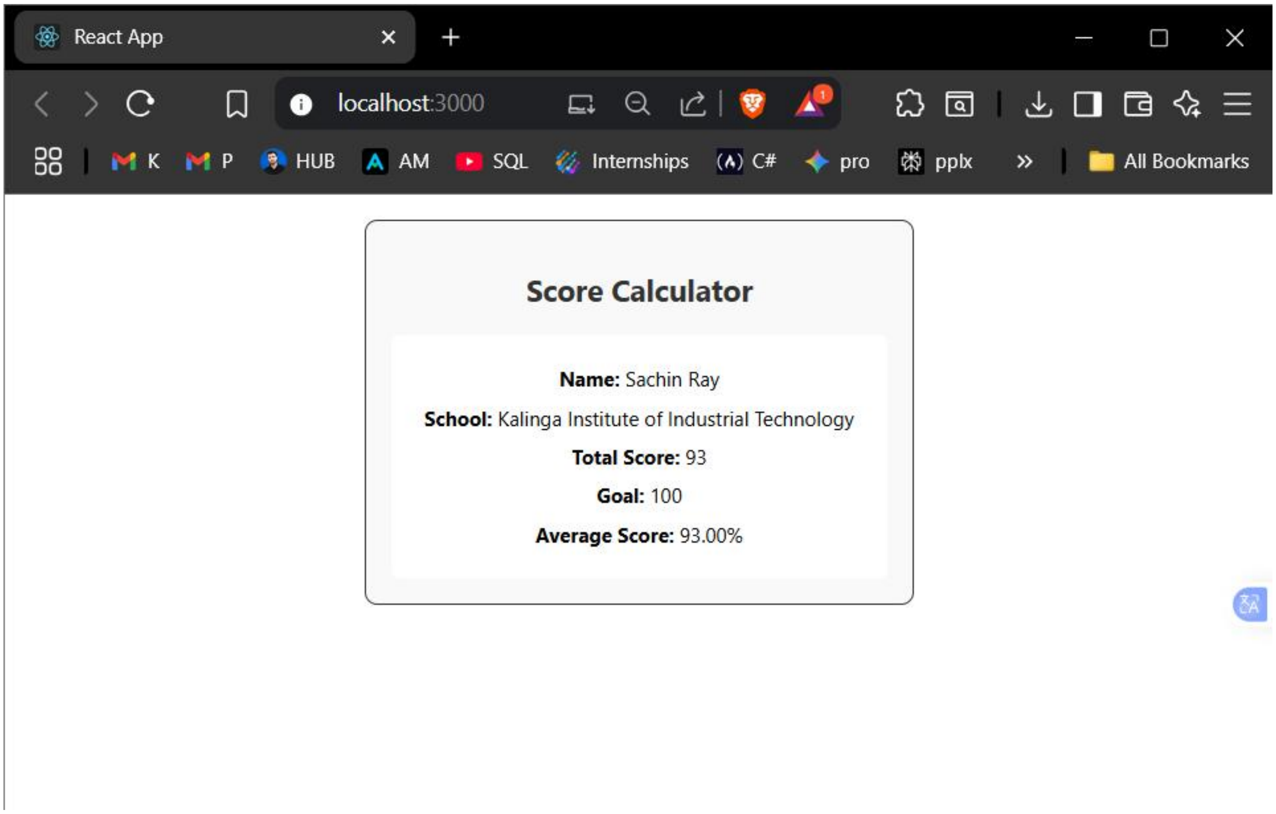

Fig. 3. LAB-3 Output

Lab 4: Component Lifecycle Hooks

Objective

Implement componentDidMount() and componentDidCatch() lifecycle hooks in a blog app.

Solution Steps

1. **Creating React project:**

```
create-react-app blogapp
cd blogapp
```

2. **Creating Post.js:**

```
// src/Post.js
class Post {
  constructor(userId, id, title, body) {
    this.userId = userId;
    this.id = id;
    this.title = title;
    this.body = body;
  }
}

export default Post;
```

3. **Creating Posts.js:**

```
// src/Posts.js
import React, { Component } from 'react';
import Post from './Post';

class Posts extends Component {
  constructor(props) {
    super(props);
    this.state = {
      posts: []
    };
  }

  loadPosts = async () => {
    try {
      const response = await fetch('https://jsonplaceholder.typicode.com/posts');
      const data = await response.json();
      const posts = data.slice(0, 10).map(post =>
        new Post(post.userId, post.id, post.title, post.body)
      );
      this.setState({ posts });
    } catch (error) {
      console.error('Error loading posts:', error);
    }
  }

  componentDidMount() {
    this.loadPosts();
  }
```

```
 componentDidCatch(error, errorInfo) {
  alert(`An error occurred: ${error.message}`);
  console.error('Error caught by componentDidCatch:', error, errorInfo);
 }

 render() {
  return (
   <div style={{ padding: '20px' }}>
    <h1>Blog Posts</h1>
    {this.state.posts.map(post => (
     <div key={post.id} style={{
      marginBottom: '20px',
      padding: '15px',
      border: '1px solid #ccc',
      borderRadius: '5px'
     }}>
      <h3>{post.title}</h3>
      <p>{post.body}</p>
     </div>
    ))}
   </div>
  );
 }
}

export default Posts;
```

4. **App.js:**

```
import React from 'react';
import './App.css';
import Posts from './Posts';

function App() {
 return (
  <div className="App">
   <Posts />
  </div>
 );
}

export default App;
```

Output:

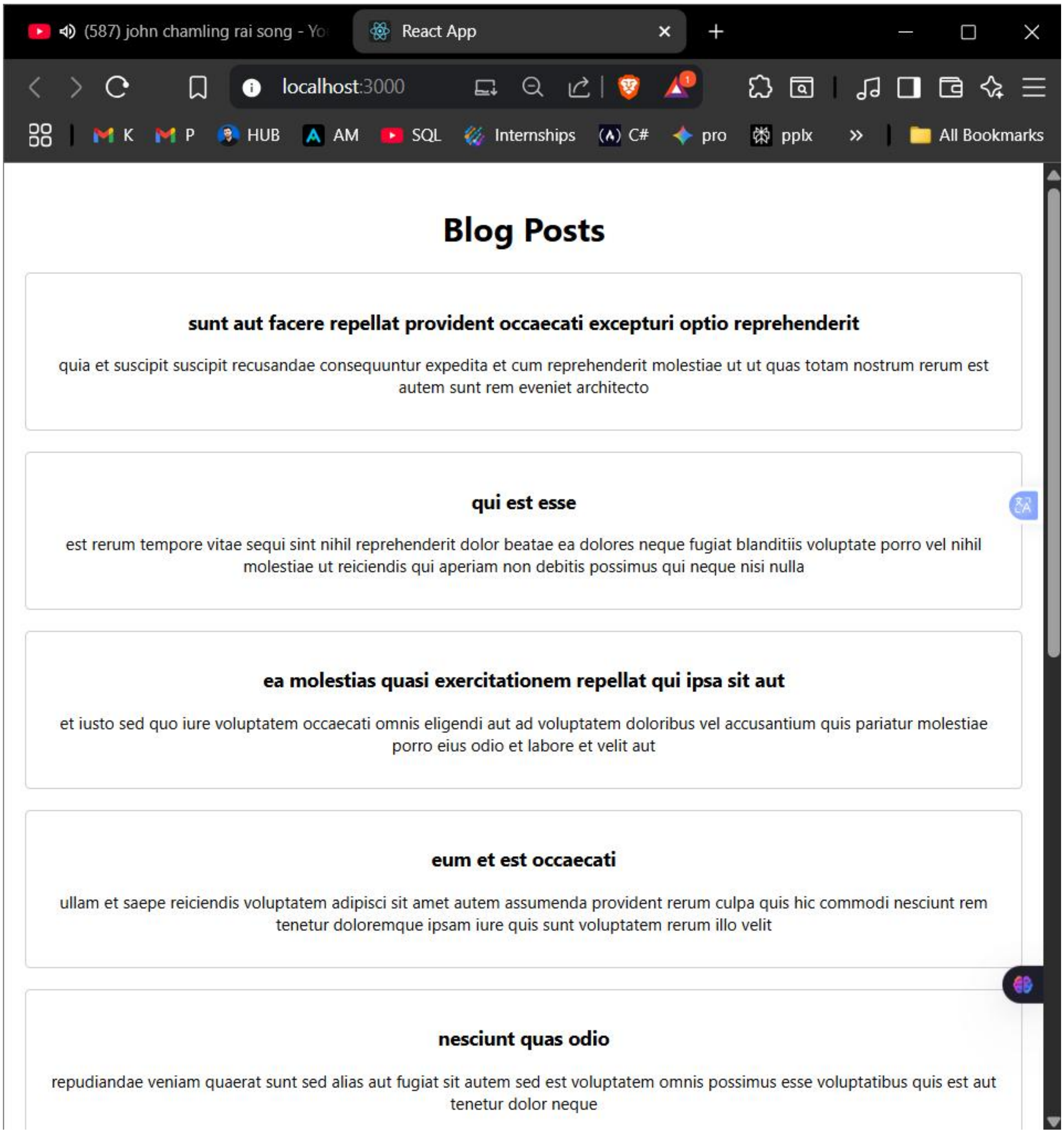

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Fig. 4. LAB-4 Output of blog-app

Lab 5: CSS Modules and Styling

Objective

Style React components using CSS modules and inline styles.

Solution:

1. **Creating CohortDetails.module.css:**

```
.box {
  width: 300px;
  display: inline-block;
  margin: 10px;
  padding: 10px 20px;
  border:2px solid black;
  border-radius: 50px;
  background-color: #f9f9f9;
}
h1{
  font-weight:1200;
  color: brown;

}
dl{
  text-align: left;
}




dt {
  font-weight: 500;
  margin-top: 10px;
}
```

2. **Creating CohortDetails.js:**

```
import React from 'react';
import styles from './CohortDetails.module.css';


const CohortDetails = ({ cohorts }) => {
  return (
    <div>
      <h1>Cohort Details</h1>
      {cohorts.map(cohort => (
        <div key={cohort.id} className={styles.box}>
          <h3 style={{
            color: cohort.status === 'ongoing' ? 'Green' : 'blue'
          }}>
            {cohort.name}
          </h3>
          <dl>
            <dt>Current Status:</dt>
            <dd>{cohort.status}</dd>
            <dt>Start Date:</dt>
            <dd>{cohort.startDate}</dd>
            <dt>Duration:</dt>
            <dd>{cohort.duration}</dd>
            <dt>Coach:</dt>
            <dd>{cohort.coach}</dd>
            <dt>Trainer:</dt>
            <dd>{cohort.trainer}</dd>
            <dt>Participants:</dt>
            <dd>{cohort.participants}</dd>
          </dl>
        </div>
      ))}
    </div>
  );
};


export default CohortDetails;
```

3. **Updating App.js:**

```
import React from 'react';
import './App.css';
import CohortDetails from './CohortDetails';
```

```
function App() {
  const cohorts = [
    {
      id: 1,
      name: 'React Fundamentals',
      status: 'ongoing',
      startDate: '2025-04-15',
      duration: '4 weeks',
      coach: 'Sachin',
      trainer: 'Bhavesh',
      participants: 25
    },

    {
      id: 2,
      name: 'Java Full-stack-Dev',
      status: 'ongoing',
      startDate: '2025-06-07',
      duration: '12 weeks',
      coach: 'Archi',
      trainer: 'Sweta',
      participants: 120
    },
    {
      id: 3,
      name: 'Advanced JavaScript',
      status: 'completed',
      startDate: '2024-11-01',
      duration: '8 weeks',
      coach: 'Himanshu',
      trainer: 'Priyank',
      participants: 75
    }
  ];


  return (
    <div className="App">
      <CohortDetails cohorts={cohorts} />
    </div>
  );
}
```

```
export default App;
```

Output:

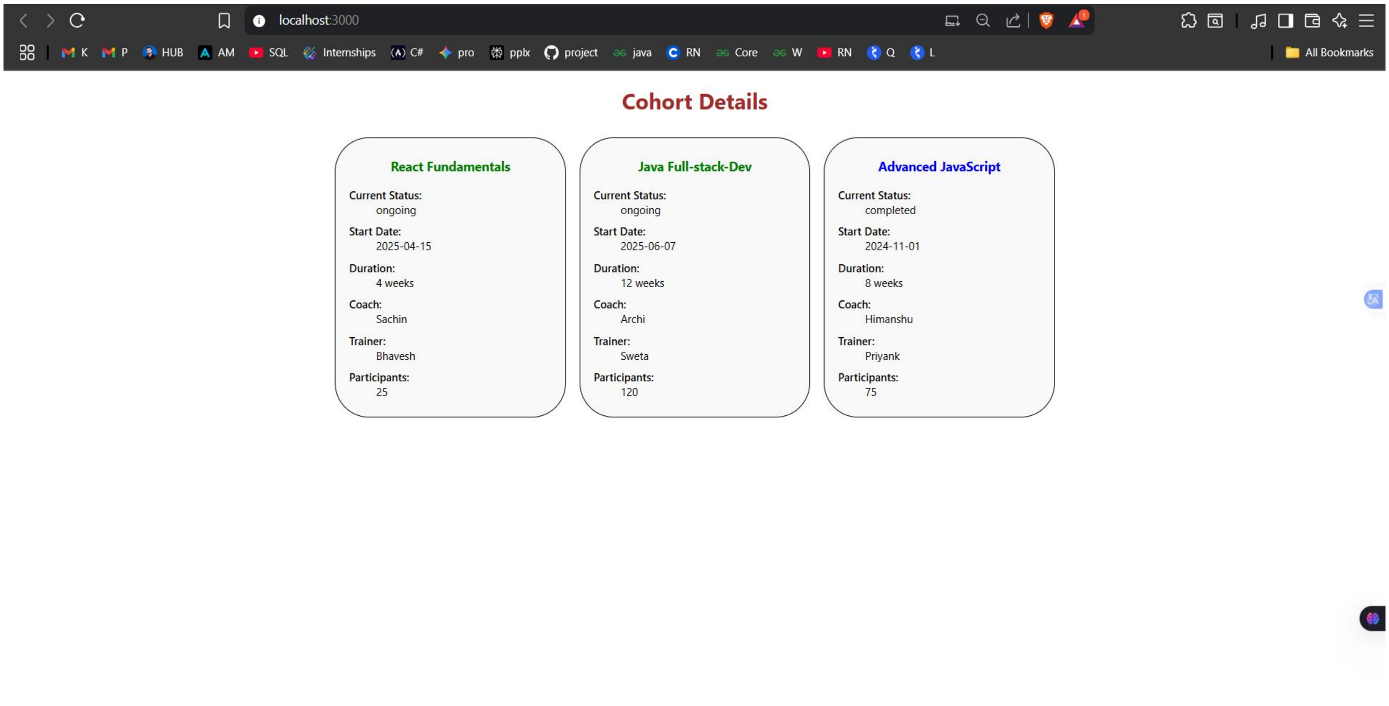


Fig. 5: LAB-5 Output of cohorttracker

ADDITIONAL HANDSON Done For Practice

Lab 6: React Router Implementation

Objective

Create a trainers app with routing functionality.

Solution:

1. **Creating React project and installing router:**

```
create-react-app TrainersApp
cd TrainersApp
npm install react-router-dom
```

2. **Creating Trainer.js:**

```
class Trainer {
  constructor(trainerId, name, email, phone, technology, skills) {
    this.trainerId = trainerId;
    this.name = name;
    this.email = email;
    this.phone = phone;
    this.technology = technology;
    this.skills = skills;
  }
}

export default Trainer;
```

3. **Creating TrainersMock.js:**

```
import Trainer from './Trainer';


const trainers = [
  new Trainer('T-001', 'Sachin Ray', 'sachin122np@email.com', '9124049407', 'React', ['JavaScript', 'React', 'Redux', 'React Native']),
  new Trainer('T-002', 'Sam ALtair', 'sammyhopes@email.com', '987654321', 'Angular', ['TypeScript', 'Angular', 'RxJS']),
  new Trainer('T-003', 'Pinaki Johnson', 'pinakijohn@email.com', '5551234567', 'Vue', ['JavaScript', 'Vue.js', 'Vuex']),
  new Trainer('T-004', 'Archi Singh', 'Archi1431@email.com', '4449876543', 'Node.js', ['JavaScript', 'Node.js', 'Express']),
  new Trainer('T-005', 'David Shetty', 'david@email.com', '3345221111', 'Python', ['Python', 'Django', 'Flask'])
];


export default trainers;
```

4. **Creating Home.js:**

```
import React from 'react';
const Home = () => {
  return (
    <>
      <div style={{ textAlign: 'center', marginTop: '50px', backgroundColor:'skyblue' }}>
```

```
      <h1>Welcome to Tech Trainers Portal</h1>

     </div>

     <div>

      <p>Manage and view trainer information</p>

     </div>

   </>

  );

};

export default Home;
```

5. **Creating TrainersList.js:**

```
import React from 'react';
import { Link } from 'react-router-dom';
import trainers from './TrainersMock';

const TrainersList = () => {
  return (
    <div style={{ padding: '20px' }}>
      <h2>Trainers List</h2>
      <ul style={{ listStyle: 'none', padding: 0 }}>
        {trainers.map(trainer => (
          <li key={trainer.trainerId} style={{ margin: '10px 0' }}>
            <Link
              to={`/trainer/${trainer.trainerId}`}
              style={{
                textDecoration: 'none',
                color: 'blue',
                fontSize: '18px'
              }}
            >
              {trainer.name}
            </Link>
          </li>
        ))}
      </ul>
    </div>
  );
};
```

```
export default TrainersList;
```

6. **Creating TrainerDetails.js:**

```
import React from 'react';
import { useParams } from 'react-router-dom';
import trainers from './TrainersMock';


const TrainerDetails = () => {
  const { id } = useParams();
  const trainer = trainers.find(t => t.trainerId === id);


  if (!trainer) {
    return <div>Trainer not found</div>;
  }


  return (
    <div style={{
      display: 'flex',
      justifyContent: 'center',
      alignItems: 'center',
      height: '75vh',
      padding: '20px',
      backgroundColor: 'skyblue'  // optional: light background for contrast
    }}>
      <div style={{
        maxWidth: '500px',
        border: '1px solid #ccc',
        padding: '50px',
        borderRadius: '50px',
        boxShadow: '0 2px 8px rgba(0,0,0,0.1)',
        backgroundColor: '#fff'
      }}>
        <h2>Trainer Details</h2>
        <p><strong>ID:</strong> {trainer.trainerId}</p>
        <p><strong>Name:</strong> {trainer.name}</p>
        <p><strong>Email:</strong> {trainer.email}</p>
        <p><strong>Phone:</strong> {trainer.phone}</p>
        <p><strong>Technology:</strong> {trainer.technology}</p>
        <p><strong>Skills:</strong> {trainer.skills.join(', ')}</p>
```

```
      </div>
    </div>
  );
};


export default TrainerDetails;
```

7. **App.js:**

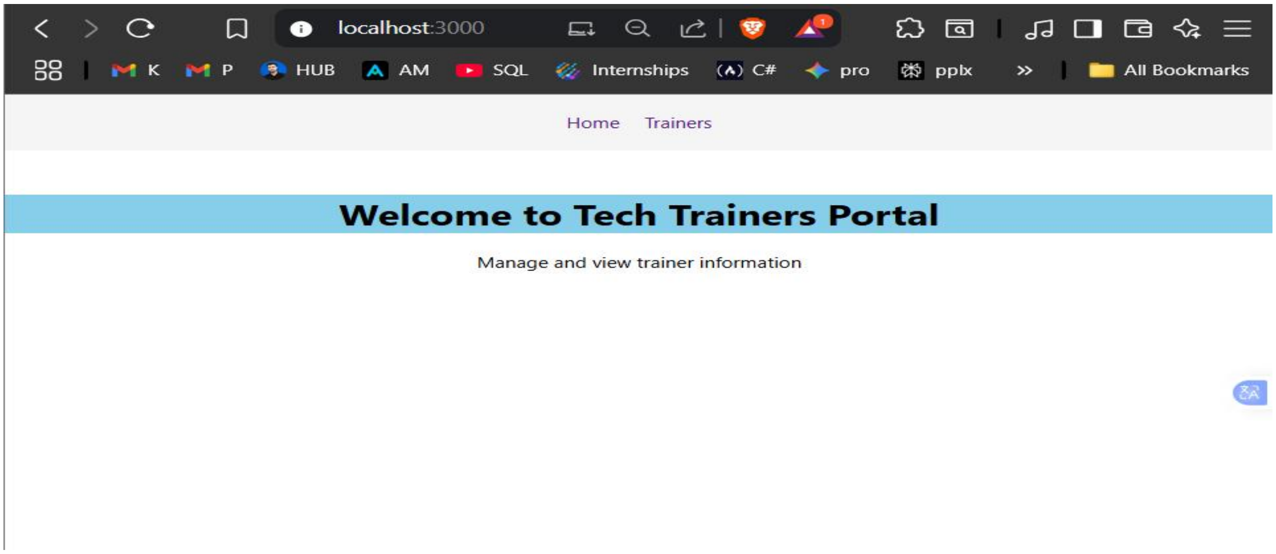
```
import React from 'react';
import { BrowserRouter as Router, Routes, Route, Link } from 'react-router-dom';
import './App.css';
import Home from './Home';
import TrainersList from './TrainersList';
import TrainerDetails from './TrainerDetails';

function App() {
  return (
    <Router>
      <div className="App">
        <nav style={{
          padding: '20px',
          backgroundColor: '#f5f5f5',
          marginBottom: '20px'
        }}>
          <Link to="/" style={{ marginRight: '20px', textDecoration: 'none' }}>
            Home
          </Link>
          <Link to="/trainers" style={{ textDecoration: 'none' }}>
            Trainers
          </Link>
        </nav>

        <Routes>
          <Route path="/" element={<Home />} />
          <Route path="/trainers" element={<TrainersList />} />
          <Route path="/trainer/:id" element={<TrainerDetails />} />
        </Routes>
      </div>
    </Router>
  );
}

export default App;
```

Output:



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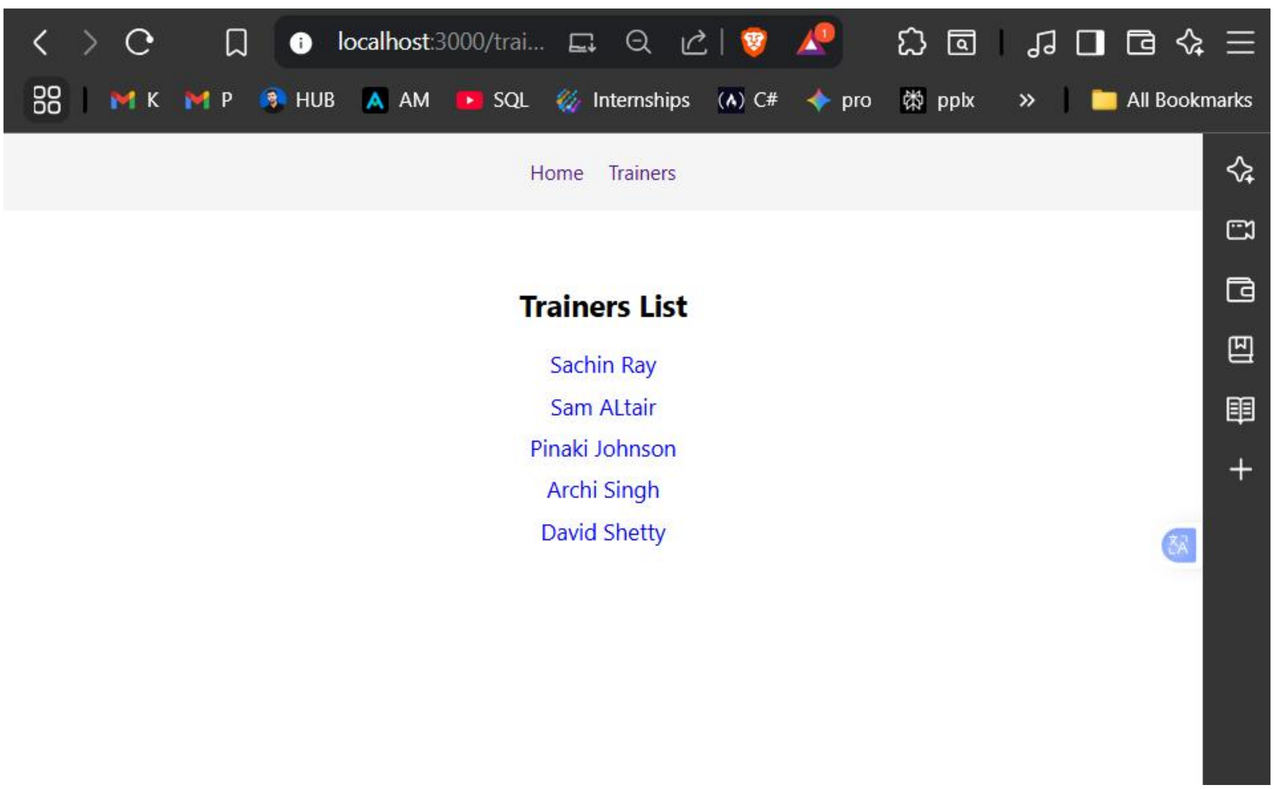
pplx
All Bookmarks
Home
Trainers
Welcome to Tech Trainers Portal
Manage and view trainer information



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Trainers List
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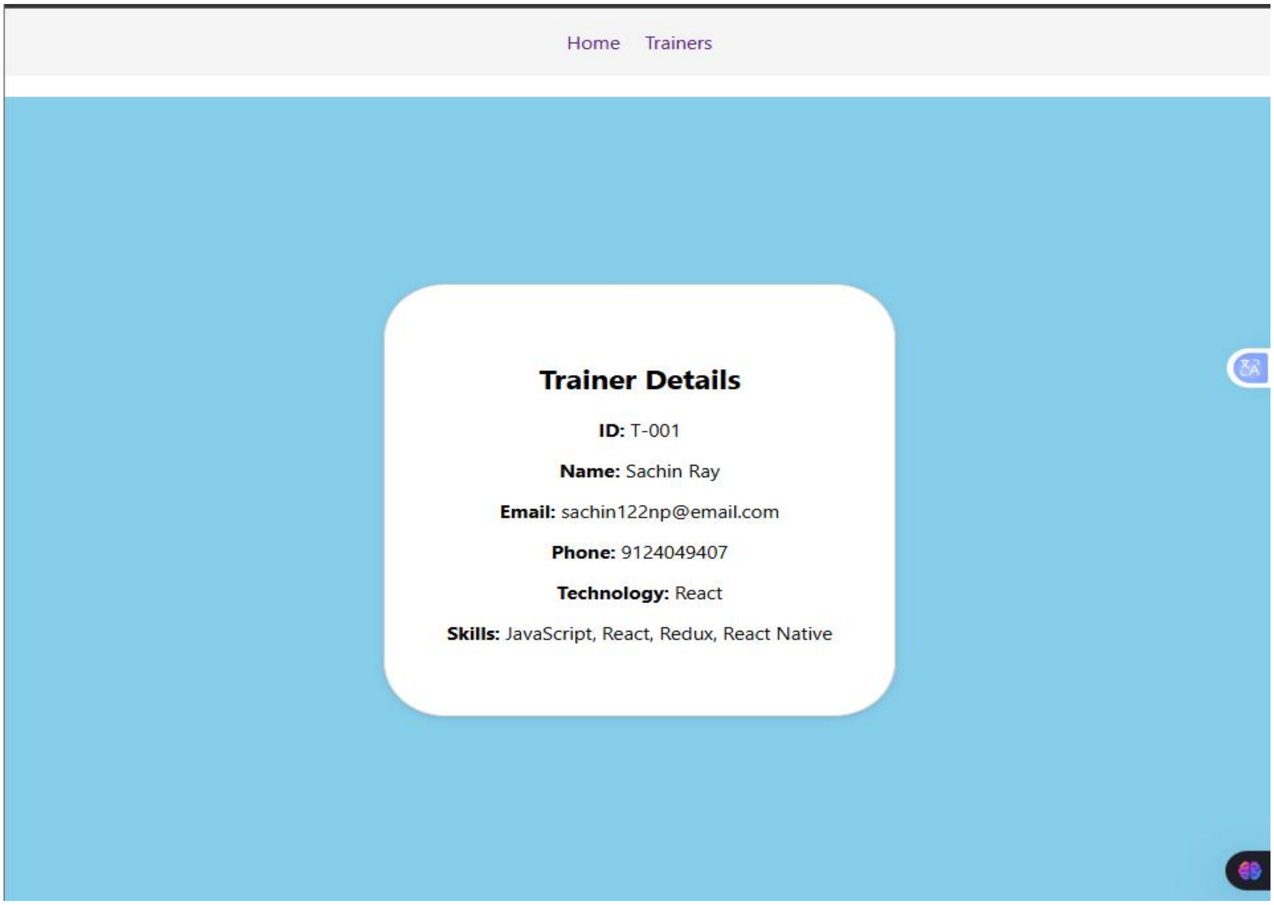
Sam ALtair
Pinaki Johnson
Archi Singh
David Shetty



Lab 7: Props Implementation

Objective

Create a shopping app using props to pass data between components.

Solution:

1. **Creating React project:**

```
create-react-app shoppingapp
cd shoppingapp
```

2. **Creating Cart.js:**

```
// src/Cart.js
class Cart {
  constructor(itemname, price, imageUrl) {
    this.itemname = itemname;
    this.price = price;
    this.imageUrl= imageUrl;
  }
}

export default Cart;
```

3. **Creating OnlineShopping.js:**

```
// src/OnlineShopping.js
import React, { Component } from 'react';
import Cart from './Cart';


class OnlineShopping extends Component {
  constructor(props) {
    super(props);
    this.cartItems = [
      new Cart(
        'Laptop',
        99999,
        'https://images.unsplash.com/photo-1517336714731-489689fd1ca8?auto=format&fit=crop&w=400&q=80'
      ),
      new Cart(
        'Mouse',
        599,
        'https://images.unsplash.com/photo-1527864550417-7fd91fc51a46?q=80&w=1167&auto=format&fit=crop&ixlib=rb-
4.1.0&ixid=M3wxMjA3fDB8MHxwaG90by1wYWdlfHx8fGVufDB8fHx8fA%3D%3D'
      ),
      new Cart(
        'Keyboard',
        2100,
        'https://images.unsplash.com/photo-1587829741301-dc798b83add3?q=80&w=1165&auto=format&fit=crop&ixlib=rb-
4.1.0&ixid=M3wxMjA3fDB8MHxwaG90by1wYWdlfHx8fGVufDB8fHx8fA%3D%3D'
      ),
      new Cart(
        'Monitor',
        43099,
        'https://images.unsplash.com/photo-1527443224154-c4a3942d3acf?w=1000&auto=format&fit=crop&q=60&ixlib=rb-
4.1.0&ixid=M3wxMjA3fDB8MHxzZWFyY2h8Mnx8TW9uaXRvcnxlbnwwfHwwfHx8MA%3D%3D'
      ),
```

```
    new Cart(
      'Headphones',
      7999,
      'https://images.unsplash.com/photo-1505740420928-5e560c06d30e?w=1000&auto=format&fit=crop&q=60&ixlib=rb-4.1.0&ixid=M3wxMjA3fDB8MHxzZWFyY2h8Mnx8aGVhZHBob25lc3xlbnwwfHwwfHx8MA%3D%3D'
    ),
  ];
}


render() {
  const totalPrice = this.cartItems.reduce((sum, item) => sum + item.price, 0).toFixed(2);


  return (
    <div
      style={{
        padding: '30px',
        fontFamily: "'Segoe UI', Tahoma, Geneva, Verdana, sans-serif",
        background: '#f5f6fa',
        minHeight: '100vh',
      }}
    >
      <h1 style={{ textAlign: 'center', marginBottom: '30px', color: '#222' }}>Online Shopping Cart</h1>
      <div
        style={{
          display: 'grid',
          gridTemplateColumns: 'repeat(auto-fit, minmax(250px, 1fr))',
          gap: '25px',
        }}
      >
        {this.cartItems.map((item, index) => (
          <div
            key={index}
            style={{
              border: '1px solid #ddd',
              boxShadow: '0 4px 8px rgba(0,0,0,0.12)',
              borderRadius: '14px',
              padding: '22px',
              backgroundColor: '#fff',
              textAlign: 'center',
              transition: 'transform 0.3s',
              cursor: 'pointer',
            }}
            onMouseEnter={e => (e.currentTarget.style.transform = 'scale(1.05)')}
```

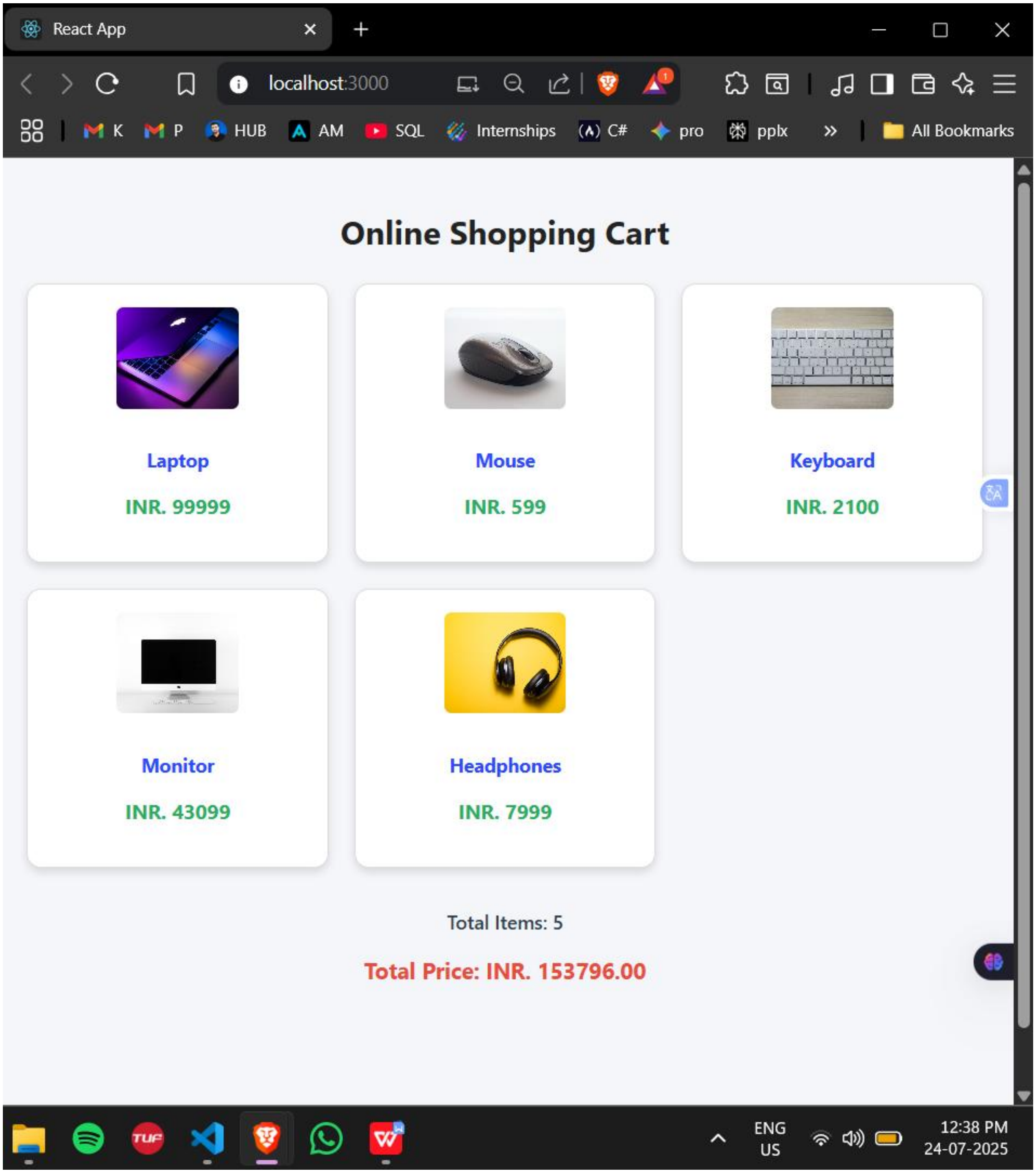
```
            onMouseLeave={e => (e.currentTarget.style.transform = 'scale(1)')}
          >
            <img
              src={item.imageUrl}
              alt={item.itemname}
              style={{
                width: '120px',
                height: '100px',
                objectFit: 'cover',
                marginBottom: '15px',
                borderRadius: '7px',
              }}
            />
            <h3 style={{ color: '#2948ff', marginBottom: '10px' }}>{item.itemname}</h3>
            <p style={{ fontSize: '20px', fontWeight: '700', color: '#27ae60' }}>INR. {item.price}</p>
          </div>
        ))}
      </div>
      <div style={{ marginTop: '40px', textAlign: 'center', color: '#2c3e50' }}>
        <p style={{ fontSize: '19px', fontWeight: '600' }}>
          Total Items: {this.cartItems.length}
        </p>
        <p style={{ fontSize: '22px', fontWeight: '700', color: '#e74c3c' }}>
          Total Price: INR. {totalPrice}
        </p>
      </div>
    </div>
  );
 }
}


export default Online
```

4. **App.js:**

```
import React from 'react';
import './App.css';
import OnlineShopping from './OnlineShopping';

function App() {
 return (
  <div className="App">
    <OnlineShopping />
  </div>
```

```
  );
}

export default App;
```

Output:



Lab 8: State Management

Objective

Create a counter app for tracking people entering and exiting a mall.

Solution Steps

1. **Create React project:**

```
create-react-app counterapp
cd counterapp
```

2. **Create CountPeople.js:**

```
// src/CountPeople.js
import React, { Component } from 'react';

class CountPeople extends Component {
  constructor(props) {
    super(props);
    this.state = {
      entrycount: 0,
      exitcount: 0
    };
  }

  updateEntry = () => {
    this.setState(prevState => ({
      entrycount: prevState.entrycount + 1
    }));
  }

  updateExit = () => {
    this.setState(prevState => ({
      exitcount: prevState.exitcount + 1
    }));
  }

  render() {
```

```
return (
  <div style={{
    textAlign: 'center',
    padding: '50px',
    maxWidth: '500px',
    margin: '0 auto'
  }}>
    <h1>Mall People Counter</h1>

    <div style={{
      backgroundColor: '#f5f5f5',
      padding: '30px',
      borderRadius: '10px',
      margin: '20px 0'
    }}>
      <h2>Current Count</h2>
      <p style={{ fontSize: '24px', margin: '15px 0' }}>
        <strong>People Entered: {this.state.entrycount}</strong>
      </p>
      <p style={{ fontSize: '24px', margin: '15px 0' }}>
        <strong>People Exited: {this.state.exitcount}</strong>
      </p>
      <p style={{ fontSize: '20px', color: 'blue' }}>
        <strong>Currently Inside: {this.state.entrycount - this.state.exitcount}</strong>
      </p>
    </div>

    <div style={{ margin: '30px 0' }}>
      <button
        onClick={this.updateEntry}
        style={{
          padding: '15px 30px',
          fontSize: '18px',
          margin: '10px',
          backgroundColor: 'green',
          color: 'white',
          border: 'none',
          borderRadius: '5px',
          cursor: 'pointer'
        }}
      >
        Entry (+1)
      </button>

      <button
```

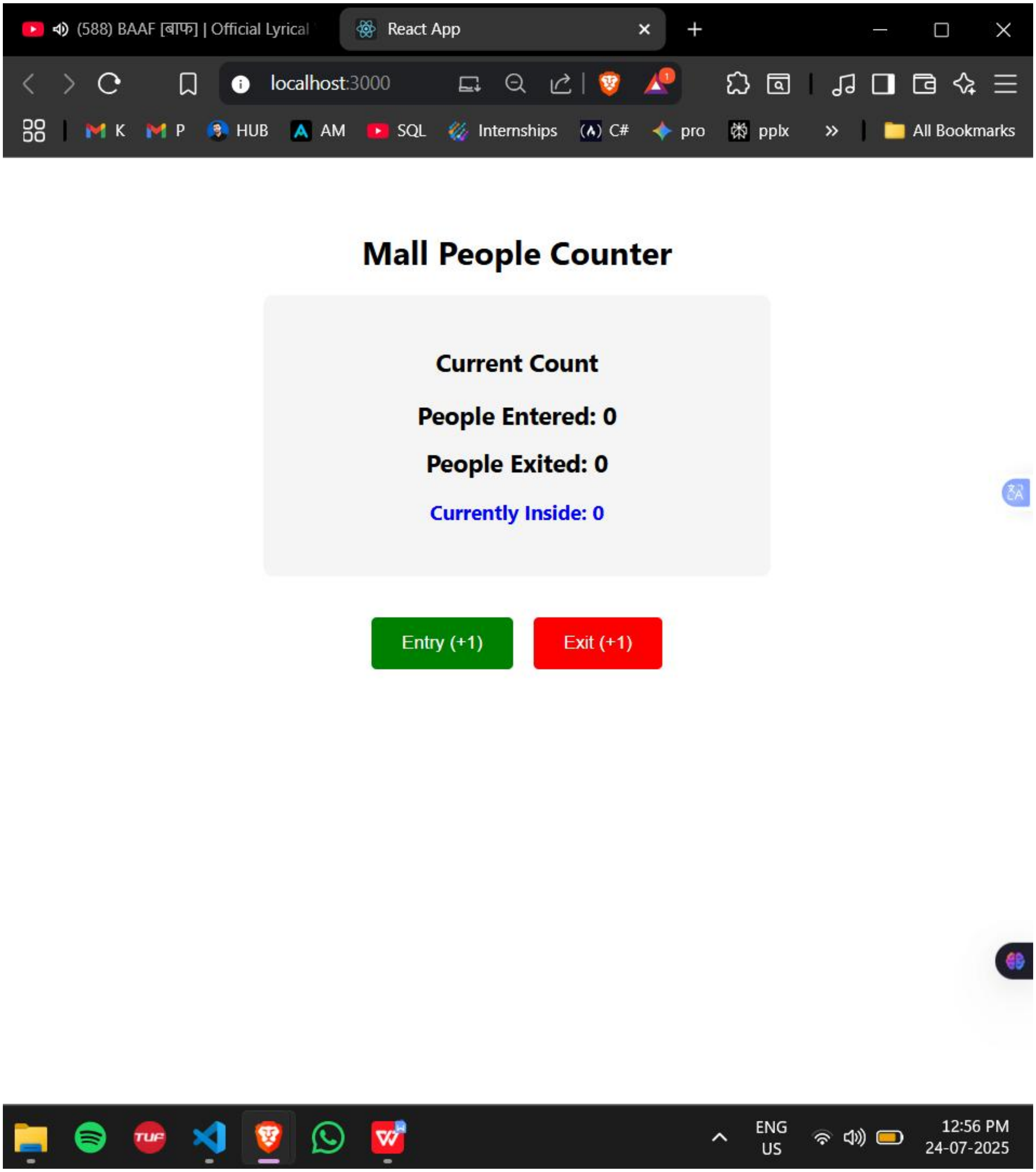
```
          onClick={this.updateExit}
          style={{
            padding: '15px 30px',
            fontSize: '18px',
            margin: '10px',
            backgroundColor: 'red',
            color: 'white',
            border: 'none',
            borderRadius: '5px',
            cursor: 'pointer'
          }}
        >
          Exit (+1)
        </button>
      </div>
    </div>
  );
 }
}

export default CountPeople;
```

3. **Update App.js:**

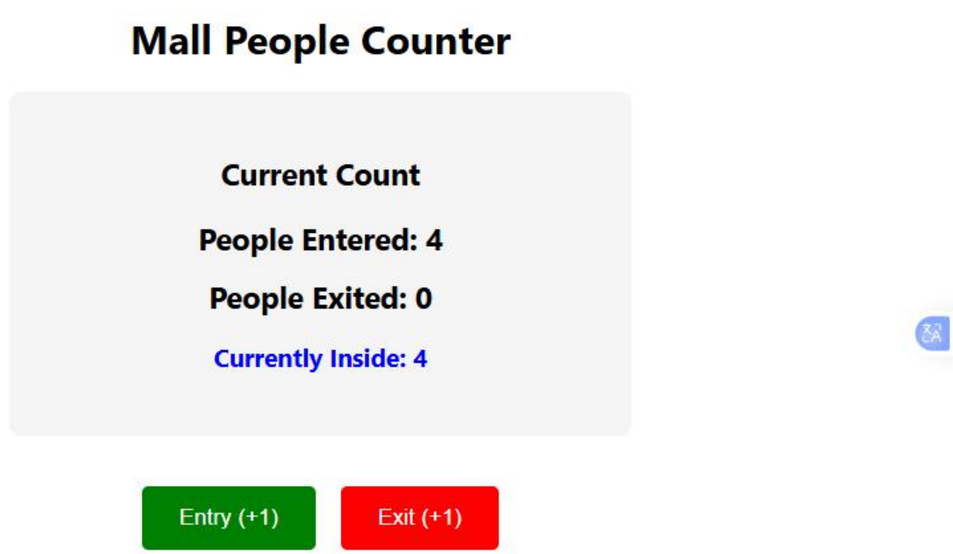
```
import React from 'react';
import './App.css';
import CountPeople from './CountPeople';

function App() {
 return (
  <div className="App">
   <CountPeople />
  </div>
 );
}

export default App;
```

Outputs:

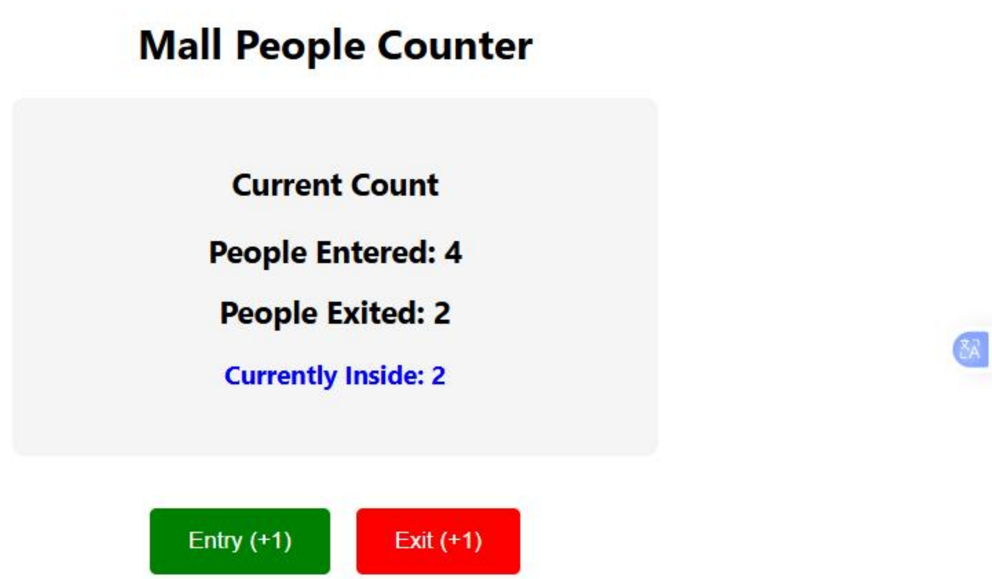
Initial State:



Entering People: when Green Clicked, State->



Exiting People: When Red Clicked, State->



DONE BY: SACHIN RAY() as a Part of >.NET FSE DEEPSKILLING ASSESSMENT PART